

PRICE PER NIGHT IN SEATTLE

Analyzing prevailing market data trends and forecast rental prices for AirBNB listings in the Seattle Metro area.





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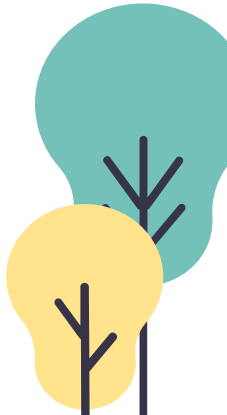
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OUR PROJECT GOALS





PROJECT GOALS

KEY FEATURES AFFECTING PRICE PER NIGHT

What are the **key features** and characteristics that make certain properties **highly sought after**, ultimately influencing their **per-night pricing**?

PERCEPTION SHAPING PRICING DYNAMICS

How does the **perception** of property reliability, as influenced by **consumer feedback** and metrics, play a role in shaping these **pricing dynamics**?

PROJECT GOALS



BACKGROUND

You want to **invest in AirBnB** real estate or want the **best deal possible**. Our central question was to find the **optimal AirBnB** to suite the **specific needs** of the businessman or consumer.



REASONING

AirBnBs are a marketplace that are very hard to find a great deal on. Because of this, it is very useful to **analyze AirBnB data** to view many different details about them throughout their history.



SIGNIFICANCE

We were able to **predict the price of AirBnBs**, analyze trends via linear regression, and view multivariable relationships through general exploratory data analysis.

DATA SOURCES & CLEANING



DATA SOURCES & CLEANING

Sourced **two** datasets
from openly
accessible **AirBNB**
archives.

Combined dataset accentuating
an array of features from over
3,000 properties in the **Seattle**
region.

Filled in **missing data**,
removed **identical**
duplicates, and dealt with
unusual data points.



STATISTICAL ANALYSIS RESULTS





STATISTICAL ANALYSIS RESULTS

ANALYSIS

$\alpha = 0.05$
 $df = 3817$

AVERAGE PRICE

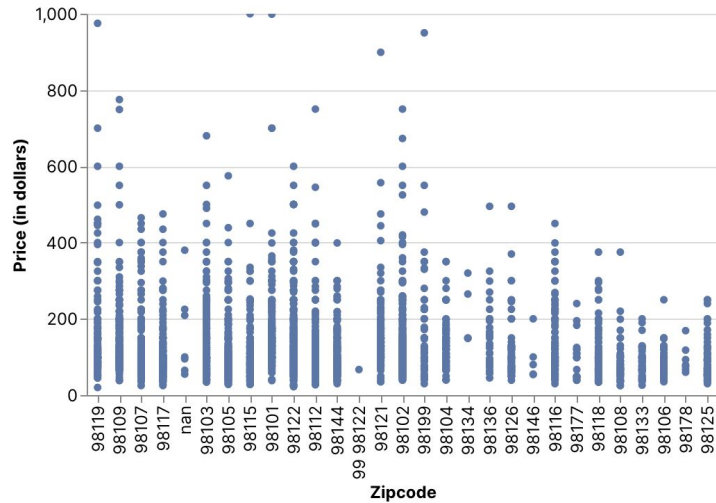
$\$127.98 \pm$
 $\$176.92$

STANDARD ERROR

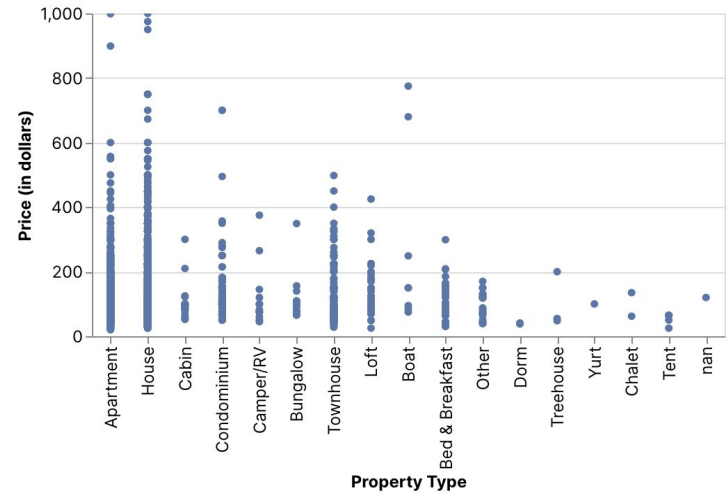
$\$90.24$

STATISTICAL ANALYSIS RESULTS

Distribution of Property Price Based on Zipcode

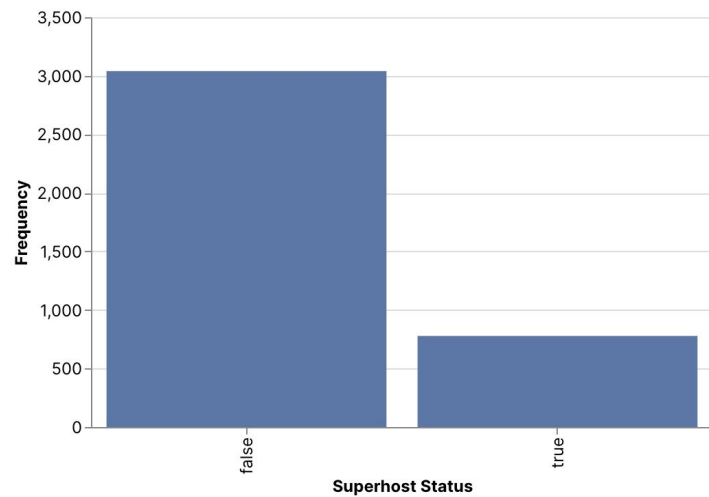


Distribution of Property Type vs. Price

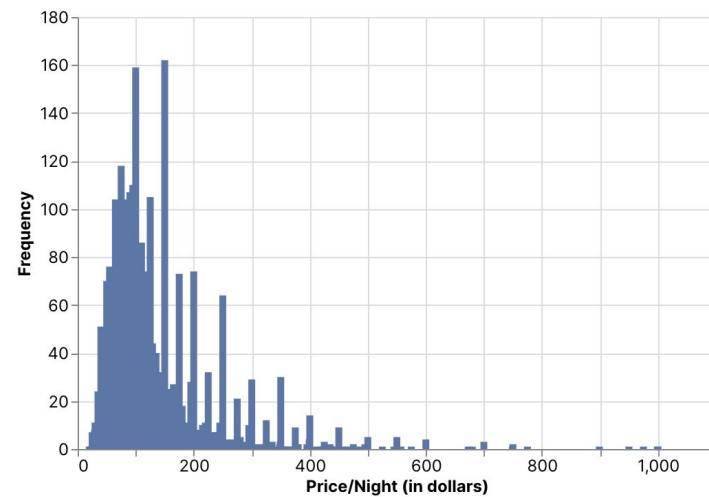


STATISTICAL ANALYSIS RESULTS

Number of Listings Posted by Superhosts vs. Non-Superhosts



Frequency Distribution of Price/Night Values for Airbnbs in Seattle

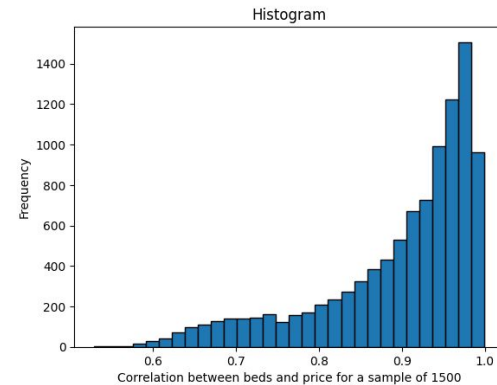
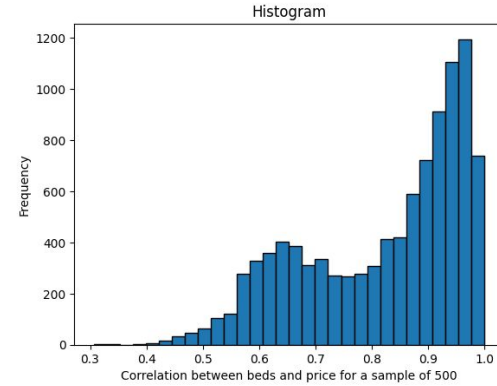
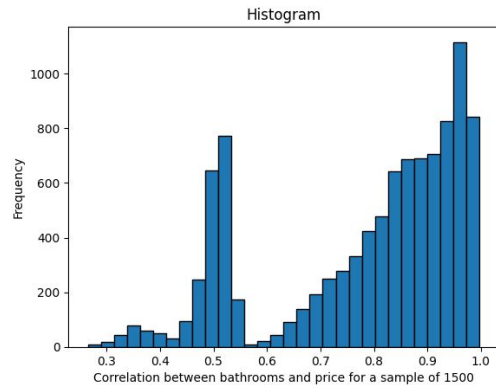
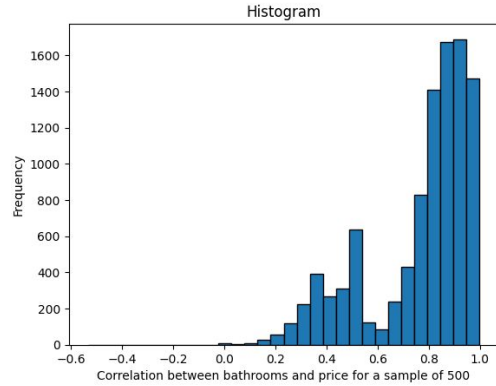


STATISTICAL ANALYSIS RESULTS

Bathrooms

VS.

Price

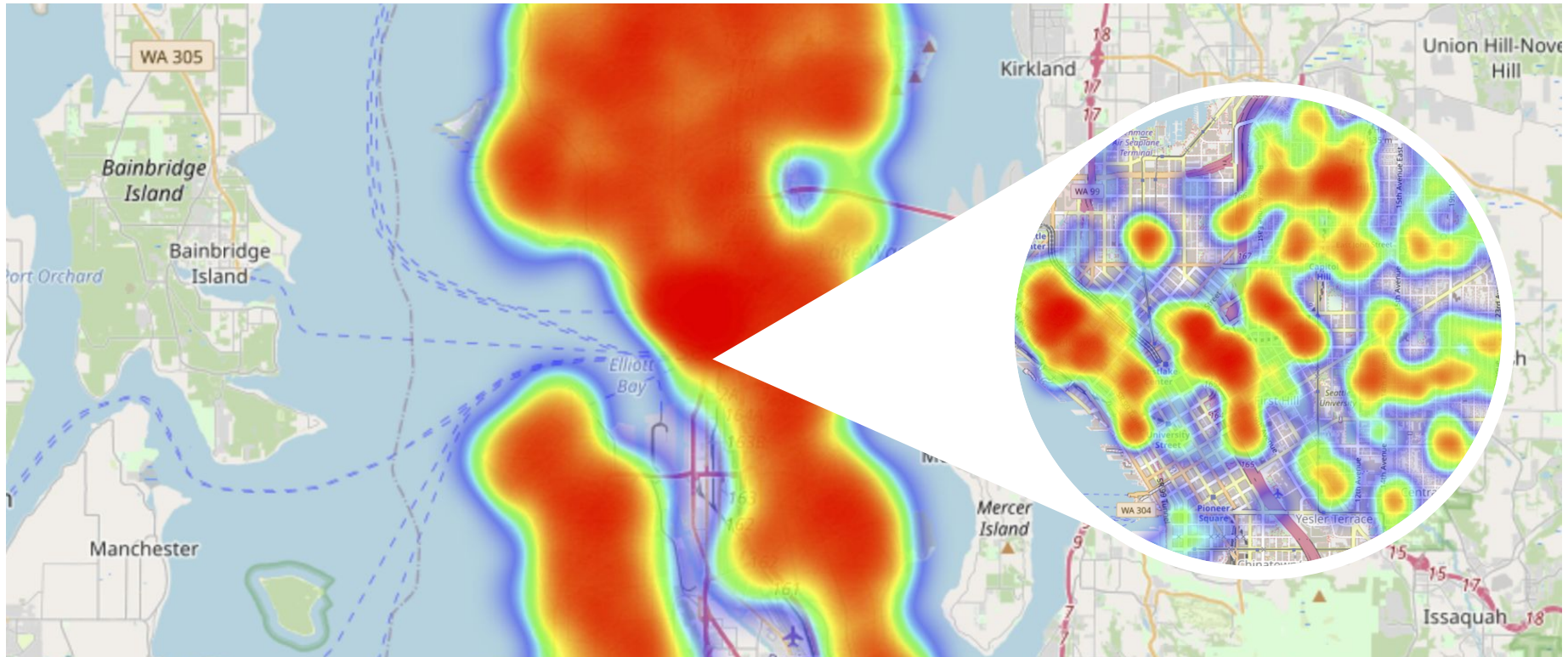


Beds

VS.

Price

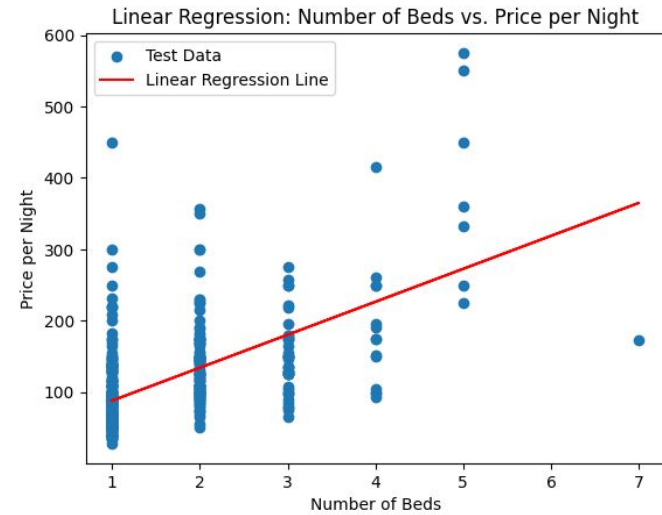
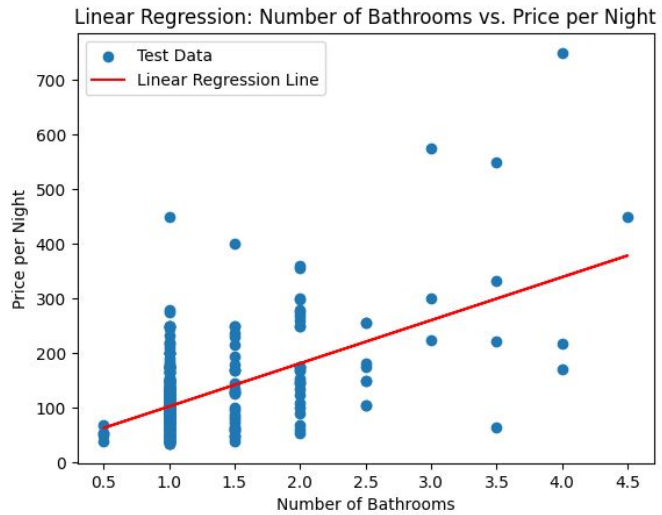
STATISTICAL ANALYSIS RESULTS



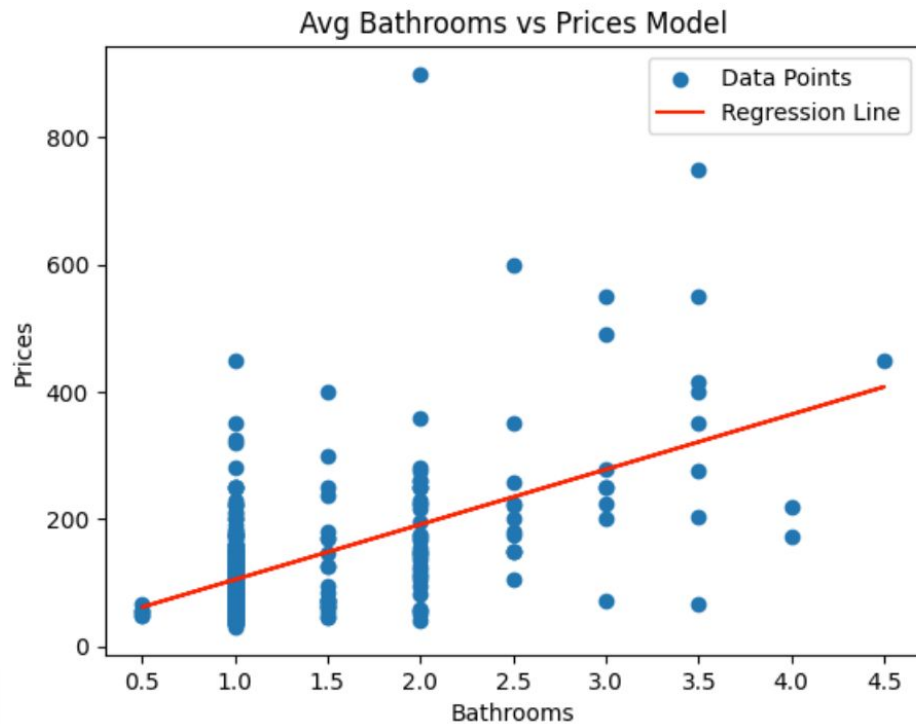
LINEAR PRICE PREDICTION MODELS



STATISTICAL ANALYSIS RESULTS



STATISTICAL ANALYSIS RESULTS



RANDOM FOREST REGRESSOR ALGORITHM





RANDOM FOREST REGRESSOR ALGORITHM

ML ALGORITHM

The Random Forest Regressor is a Machine learning algorithm that **enhances regression predictions.**

DECISION TREES

It builds multiple decision trees through **bootstrapping**, utilizing random subsets of the data and features.

PREDICTIONS

Decision trees are **used to make predictions** by averaging outputs, reducing overfitting, and improving accuracy.



RANDOM FOREST REGRESSOR ALGORITHM

Host_listings_count: 5
Accommodation_capacity: 10
Bathrooms: 5
Bedrooms: 4
Beds: 7
Security_deposit: 45
Cleaning_fee: 50
Guests_included: 7
Price_per_head_over_max_capacity: 25
Minimum_nights: 1
Maximum_nights: 10
Availability_365: 156
Number_of_reviews: 500
Review_scores_rating: 8
Calculated_host_listings_count: 20
Reviews_per_month: 10

```
graph.render("decision_tree")

# Initialize an empty dictionary to store user inputs
user_inputs = {}

# Prompt the user to input values for each feature
for feature in features:
    user_input = input(f"Enter the value for {feature}: ")
    user_inputs[feature] = float(user_input) # Assuming the input is numeric

# Create a DataFrame with the user's inputs
user_data = pd.DataFrame([user_inputs])

# Make a price prediction
predicted_price = model.predict(user_data)

# Display the predicted price
print(f"Predicted Price: ${predicted_price[0]:.2f}/night")
```

Predicted Price: \$473.28/night

FINAL TAKEAWAYS



FINAL TAKEAWAYS

